Arnab Kundu

Postdoc researcher, Age: 28 years

Simion Stoilow Institute of Mathematics
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Research Interests: Arithmetic Geometry and Motivic Homotopy Theory.

Employment

Sept 2025- **Postdoctoral Researcher**, *Simion Stoilow Institute of the Romanian Academy (IMAR)*, Bucharest, Romania, in the group of Olivier Schiffmann.

Sept 2023 – Aug 2025 **Post-Doctoral Fellow**, *University of Toronto*, Toronto, Canada, in the group of Elden Elmanto.

Education

Sept 2020 – Aug 2023 **PhD in Mathematics**, *Université Paris-Saclay*, Orsay, France, under the supervision of Kęstutis Česnavičius.
Thesis: Torsors on Smooth Algebras over Valuation Rings (Official Version)

Sept 2018 – Aug 2020 **Master : Mathématiques Fondamentales**, *Sorbonne Université*, Paris, France, Score - 18.9/20.
Mention: Très Bien

Aug 2015 – July 2018 **B.Sc.(Hons.)**, *Chennai Mathematical Institute*, Chennai, India, *CGPA - 9.83*.
Mathematics and Computer Science Undergraduate

Preprints

arXiv link **Reductification of parahoric group schemes**, *available on arXiv*, 2026.

arXiv link **$\mathbb{A}^1$-connectivity of motivic spaces**, *submitted*, 2025, with Tess Bouis.

arXiv link **Beilinson–Lichtenbaum phenomenon for motivic cohomology**, *submitted*, 2025, with Tess Bouis.

arXiv link **Isotropic torsors on smooth algebras over Prüfer rings**, *submitted*, 2025.

arXiv link **Gersten's injectivity for smooth algebras over valuation rings**, *submitted*, 2024.

Publications

jrms:2023-038... **Tilting Correspondences of Perfectoid rings**, *J. Ramanujan Math. Soc.* ***38****, No. 1 (2023) 23–32.*

Thesis

Official Version **Torsors on smooth algebras over valuation rings**, *Thesis*.

Achievements:

- Received **FSMP** (Fondation Sciences Mathématiques de Paris) PSGM Masters 2018-2020 scholarship.
- Received a **Gold Medal** for the best undergraduate academic record of the 2018 batch.
- Attended the Indian **IMO training camp** in '13, '14 and '15 at HBCSE, Mumbai.

Mathematical Visits

Spring 2026 **LMSI, Mumbai**, *Thematic Programme on Rational Points, Algebraic Cycles and the Local-Global Principle*.

Fall 2025 & 2024 **CMI, Chennai**, *invited by Vikraman Balaji*.

Fall 2024 **UBC, Vancouver**, *invited by Sujatha Ramdorai*.

Spring 2023 **KTH, Stockholm**, *invited by David Rydh*.

Fall 2022 **TU Darmstadt**, *invited by Timo Richarz*.

Colloquium Talks

November 2025 **Mathematics Colloquium**, *IMSc*, Chennai, Motivic Cohomology in mixed-characteristic.

October 2025 **Mathematics Seminar**, *CMI*, Chennai, Motivic Cohomology in mixed-characteristic.

Conference Talks

January 2026 **Motives in Montpellier**, *Montpellier, France*, How to apply motivic cohomology in p-adic Hodge theory?.

September 2025 **Algebraic and Arithmetic Geometry Week(AAG 2025)**, *Iași, Romania*, Grothendieck–Serre Conjecture for Parahoric Group Schemes.

November 2024 **Special Session on Algebraic Geometry**, *CMS Meeting*, Vancouver, Motivic Cohomology in mixed-characteristic.

June 2024 **K-theory and Brauer Groups**, *Oaxaca*, $\mathbb{A}^1$-invariance of Motivic Cohomology, (video).

December 2023 **Special Session on Homotopy Theory**, *CMS Meeting*, Montréal, Gersten's injectivity in the non-Noetherian world.

Seminar Talks

February 2026 **Seminar: Thematic Programme on Rational Points, Algebraic Cycles and the Local-Global Principle**, *LMSI, Mumbai*, Purity of Motivic Cohomology in mixed-characteristic.

- **Seminar Variétés Rationnelles**, *Paris*, Reductification of Parahoric Group Schemes.

January 2026 **Number Theory Seminar**, *Caen, France*, Purity of Motivic Cohomology in mixed-characteristic.

November 2025 **Mathematics Seminar**, *CMI*, Chennai, $\mathbb{A}^1$-connectivity of Motivic Spaces.

February 2025 **Topology Seminar**, *Columbia University*, New York, Beilinson–Lichtenbaum Conjecture in Mixed-characteristic.

January 2025 **Algebraic Geometry Seminar**, *Toronto*, Motivic Cohomology in mixed-characteristic.

December 2024 **Algebraic Geometry Seminar**, *CMI*, Chennai, Grothendieck–Serre Conjecture for Parahoric Group Schemes.

November 2024 **UBC Algebraic Geometry Seminar**, *Vancouver*, Motivic Cohomology in mixed-characteristic.

October 2024 **Number Theory Seminar**, *Toronto*, Grothendieck–Serre Conjecture for Parahoric Group Schemes, (video).

November 2023 **Algebraic Geometry Seminar**, *Toronto*, Gersten's injectivity in the non-Noetherian world.

- **Arithmetic and Algebraic Geometry Seminar**, *Allahabad*, Gersten's injectivity in the non-Noetherian world.

April 2023 **Algebra and Geometry Seminar**, *Stockholm*, Tilting Correspondences of Perfectoid Rings.

January 2023 **Algebraic Geometry Seminar**, *Purdue*, Torsors on Valuation Rings.

- **Algebra and Geometry Seminar**, *Stockholm*, Torsors on Valuation Rings.
- **Stacks Seminar**, *Essen*, Torsors on Valuation Rings.
- **Arithmetic and Algebraic Geometry Seminar**, *Amsterdam*, Torsors on Valuation Rings.

December 2022 **Arithmetic Geometry Seminar**, *Heidelberg*, Torsors on Valuation Rings.

- **Seminar on Arithmetic Geometry**, *Darmstadt*, Torsors on Valuation Rings.

November 2022 **Preprint Seminar**, *Paris*, Torsors on the Complement of a Smooth Divisor.

October 2022 **Seminar Mathjeunes**, *Paris*, Torsors on Valuation Rings.

June 2022 **Seminar SAGA**, *Orsay*, Tilting Correspondences of Perfectoid Rings.

December 2021 **Seminar RéGA**, *Paris*, Tilting Correspondences of Perfectoid Rings.

Organisation of Seminar

2024-2025 **Algebraic Geometry Seminar**, *University of Toronto*, Toronto, Canada, with E. Elmanto, Daniel Litt, Nick Rozenblyum, Sebastián Olano, Website.

- **Number Theory Seminar**, with Florian Herzig, Website.

Fall 2024 **Seminar on the Unramified Cohomology**, with E. Elmanto.

Fall 2023 **Seminar on the Grothendieck–Serre Conjecture**, with E. Elmanto, Website.

2022-2023 **Seminaire Réseau des étudiants en Géométrie Algébrique (RéGA)**, *Institut Henri Poincaré (IHP)*, Paris, France, Website.

Teaching Experience

Spring 2025 & Fall 2023 **MAT223 Linear Algebra I**, *Course Instructor*, University of Toronto, Canada.

Spring 2025 **MAT224 Linear Algebra II**, –.

Spring 2024 **MAT344 Intro to Combinatorics**, –.

Fall 2022, 2021 & 2020 **Mathematics - 1**, *Teacher*, Institut Universitaire de Technologie (IUT). Cachan, France

Fall 2016 **Analysis 1**, *Teaching Assistant*, Chennai Mathematical Institute (CMI), Chennai, India.

January 2016 **Indian National Mathematical Olympiad Training Camp**, *Instructor*, The Institute of Mathematical Sciences (IMSc), Chennai.

Workshops and Conferences Attended

Spring 2024 **Workshop on p-adic Arithmetic Geometry (Spring)**, *IAS*, Princeton, USA.

– **Workshop on p-adic Arithmetic Geometry (Fall)**, *IAS*, Princeton, USA.

Summer 2023 **Stacks Project Workshop 2023**, *University Of Michigan*, Ann Arbor, USA.

– **Summer School 2023**, *University of Wuppertal*, Wuppertal, Germany.

– **Recent Advances in Algebraic K-Theory**, *IHES*, Bures-sur-Yvette, France.

Spring 2023 **International Workshop on Motives**, *University of Tokyo*, Tokyo, Japan.

Spring 2022 **Franco-Asian Summer School on Arithmetic Geometry**, *CIRM*, Luminy, France.

– **Géométrie Algébrique en Liberté (GAeL) XXIX**, *Université Paris-Saclay*, Orsay, France.

– **Cohomology of Varieties**, *IMPAN*, Warsaw, Poland.

– **Kleine AG: Weil Conjectures**, *Universität Bonn*, Germany.

Fall 2019 **Perfectoid Spaces**, *ICTS*, Bangalore, India.

Summer 2019 **Arithmetic Geometry in Carthage**, *Tunisian Academy Beit al-Heikma*, Carthage, Tunisia.

Summer 2018 **École Jeunes Chercheurs en Théorie des Nombres 2018**, *Université de Franche-Comté Laboratoire de Mathématiques de Besançon*, Besançon, France.

Winter 2017 **ATMW Analytic Number Theory**, *ISI*, Kolkata, India.

Winter 2016 **Theoretical and Computational Aspects of the Birch and Swinnerton-Dyer Conjecture**, *ICTS*, Bangalore, India.

Languages

Bengali (Native), English, French, Hindi German (Learning)

Computer skills

Programming Python, Haskell, C++, Java Web design HTML, CSS